Going in depth: the five questions, answered from the ground up

Jude's structure for tomorrow: what was working before, how it was working, what I built, how that works, and the results before and after. This is my study copy. Every idea is explained from zero, so I can answer follow-ups, not just recite.

Nada. For the meeting on 4 October 2026.

0. The foundations everything else stands on

Read this part until it feels obvious. Every answer tomorrow is built from these pieces.

A fingerprint. PDQ turns one picture into 256 ones and zeros, a tiny summary of where the light and dark areas sit in the picture. Two pictures of the same thing give nearly the same summary, even after a filter or some compression.

Distance. To compare two fingerprints, count how many of the 256 bits differ. That count is the distance. Zero means identical. Under about 20, it is the same picture for all practical purposes. 21 to 35, probably the same picture, maybe filtered or cropped. 36 to 44, might be the same, might be a look-alike. 45 and up is thrown away as noise.

The library (the index). Every film we protect is fingerprinted in advance, about three frames per second of film. My laptop copy holds 134 films, 16.7 million fingerprints. Each stored fingerprint remembers which film it came from and at what time in the film it sits.

A query. When a clip comes in, the engine fingerprints its frames and asks the library: for each clip fingerprint, which stored fingerprints are closest? Every answer under distance 45 is one piece of evidence: "this frame of the clip looks like film X at minute M".

The offset. For each piece of evidence, subtract the clip time from the film time. If my clip really is a scene from the film, every frame matches the film at the same offset: clip second 1 matches film minute 40:01, clip second 2 matches 40:02. Same offset all the way. A coincidence scatters: a stray hit at minute 10, another at minute

50. The engine's own documentation calls the constant offset the signature of a true

match. Remember this one; it is the heart of the whole week.

1. What was working before

The pod's PDQ engine, which I have as an exact laptop copy (verified identical). It has three parts that can produce an answer, and they run in this order:

1. **The pre-scan.** Before anything else, it grabs 40 frames spread across the whole

clip and does a quick vote. If one film clearly owns those pictures, it answers right there and nothing else runs.

2. **The probe (rule P0).** If the pre-scan didn't answer, the engine starts reading the

clip from the beginning, and after about a second it takes one quick look: is there any single frame that looks close to some film? If yes, answer on the spot.

3. **The rules (P1 to P9).** If neither quick part answered, the engine keeps reading

the clip, builds a scorecard per candidate film, and a list of rules watches those scorecards. First rule to fire wins. Some rules want a few excellent frames, some want piles of mediocre ones, the careful ones also check the offsets.

And it genuinely was working, mostly: on my test set of 125 clips with known answers, it identified 52 clips correctly. It also gave 7 wrong answers, the thing this whole project is about.

2. How it was working, exactly

The pre-scan's vote. Each of the 40 sampled frames counts as a "hit" on a film if its closest stored fingerprint is under distance 28. A film is accepted if it collects at least 3 hits and the best one is under 20. That's the whole test. It never looks at offsets, never asks whether the hits agree about where in the film they land, never asks whether they came from different parts of the clip. Count and best, nothing else.

The probe's test. In the first seconds of the clip it collects up to 64 fingerprints from one crop shape, searches once, and accepts if any single fingerprint is under distance 30. One frame. 30 is in the "probably" band, not the "certain" band.

The rules. The scorecard for each film counts matched frames in three quality bands, remembers the best distance, and keeps two kinds of histogram: where in the film the hits land (the offsets), and which seconds of the clip produced them. The careful rules use all of it. P4, for instance, wants five strong frames that agree on film position AND come from two different moments of the clip. But the rules only ever see the clips the quick parts passed over.

Where the wrong answers came from. I ran all 125 clips and recorded which part produced every answer. The seven wrong ones:

part	wrong answers	the clips
pre-scan	4	Krrish 3 (6), The Northman (6), Civil War (12), The In Between (18)
probe P0	1	Smile (26.8, five frames)
rule P9E	1	Grave Encounters 2 (two frames, avg 34)
rule P8	1	Demeter (31 weak frames, avg 41)

The numbers in brackets are distances. Look at the pre-scan four: distances 6 to 18. Those pictures genuinely are in our library, attached to the wrong film, almost certainly shared footage like trailers. And because the pre-scan answers before the rules run, no rule change can ever touch them. The last three

are the rules I'd already tightened in previous weeks (P8 and P6, then P9E and P0 designed).

So the structural problem: the engine's checking and its deciding live in different places. The parts that decide most of the traffic (pre-scan, probe) have the least checking. The careful checks exist, in the mid-tier rules, but they rarely get to decide anything.

3. What I built

Four things, each small on its own.

First, attribution testing (`attribute_rules.py`). Every clip run now records which part of the engine gave the answer. Without this, "seven wrong answers" is a number; with it, every wrong answer has the name of the rule that made it next to it. The table in section 2 is its output. This is also how every later change gets judged.

Second, the rebuilt probe (`pdq_engine_PROBEV2.py`). One decider that has both the quick parts' speed and the careful rules' checks:

- It samples 40 frames across the whole clip, like the pre-scan.
- For each sampled frame it fingerprints every view the full engine knows: the full

frame, the caption-trimmed frame, and all the crop shapes. The old pre-scan and probe each used a subset.

- Per film it keeps a mini scorecard: how many distinct frames hit it, the best

distance, and for every hit the pair (clip time, film time), so it can check the offsets.

- It answers only in three situations, each copied from a rule the pod already trusts:
- **PV-A**: two near-identical frames (copied from P2.5)
- **PV-B**: the old pre-scan bar, PLUS at least two hits agreeing on the offset

(pre-scan + P4's check)

- **PV-C**: five medium frames agreeing on offset, from two different moments of the

clip (copied from P6a)

- The old one-frame P0 is retired; this replaces it. The streaming rules stay behind it

as a backup for clips the probe isn't sure about.

Third, the edits harness (`make_edited_clips.py`, `test_edits.py`). Eight clips the engine identifies correctly, ten edited copies of each: filter, black and white, recompression, speed up, slow down, zoom, slight rotation, caption bars, letterbox, mirror. Since the originals work, any failure is the edit's fault. It found the engine completely blind to mirrored clips (0 of 8) and letterboxed clips (1 of 8).

Fourth, the two blindness fixes, both inside the probe:

- **Mirror**: the fingerprint describes where things sit in the picture; flip the

picture and the fingerprint comes out completely different, and nothing in the engine ever looked at the flipped version. The probe now also fingerprints the mirror of every view it samples.

• **Letterbox**: the engine fingerprinted the whole frame, which is mostly black border

once the clip is shrunk inside bars. The pod has an "active region crop" that finds the picture inside the borders; it was one of the pieces never ported to the laptop copy. I built a simple version: detect the near-black rows and columns, cut them off, fingerprint what's inside.

4. How the new probe works, on one worked example

Take a clip that really is a scene from Soul.

1. The probe grabs 40 frames spread across the clip. Frame at clip second 3 gets

fingerprinted as the full frame, the trimmed frame, the crops, and the mirror of each.

2. Each fingerprint asks the library for its closest neighbours. Suppose the second-3

frame's best answer is "Soul, film minute 41:03, distance 11".

3. The probe writes that on Soul's mini scorecard with the pair (clip 0:03, film

41:03). Offset: 41 minutes.

4. The same happens for the other frames. Clip 0:10 matches film 41:10. Clip 0:21

matches 41:21. Same offset, 41 minutes, every time.

5. Now the bars. Best distance 11 isn't under 12 with a second frame under 16, so not

PV-A. But: three or more frames under 28, best under 20, and at least two agreeing on the offset? Yes, easily, they ALL agree. PV-B fires. Answer: Soul.

Now the same mechanics on a look-alike. A clip wrongly matching The Northman has one frame at distance 6 (shared footage), and a few others at 16 to 28 scattered across the film: offsets 12 minutes, 55 minutes, 80 minutes. Count and best look fine, which is why the old pre-scan answered Northman. But no two hits agree on the offset, so PV-B refuses, and PV-A needs two near-perfect frames and there's only one. The probe stays quiet. That's the entire trick: real scenes agree, coincidences scatter.

The one case the trick can't solve: when the clip's pictures are identical to library pictures at distance 6 AND agree (Krrish 3), there is nothing left for picture evidence to say. And the one real cost: fan edits of a true film jump between scenes, so their offsets genuinely scatter, and the strict probe declines four of those. They look, on pictures alone, exactly like the Northman case. That's the policy question I'm bringing.

5. Results, before and after

On the real clips (125 with known answers):

	wrong answers	correct answers
before: the pod's rules	7	52
after: new probe + rules as backup	**1**	**48**
(new probe alone, everything else off)	1	41

Per wrong answer: Demeter (P8), Grave Encounters 2 (P9E), Smile (P0) are stopped by the rule changes and the probe replacing P0. Northman, Civil War, In Between are refused by the agreement check. Krrish 3 remains, identical pictures, needs audio or SSCD. The four correct answers lost are the fan-edit clips above.

On the edited clips (8 per edit):

edit	before	after
mirror	0 of 8	**8 of 8**, at distance 2
letterbox	1 of 8	**6 of 8** (two have bars that aren't quite black)
filter, b&w, recompression, speeds	6-8 of 8	unchanged
rotation, zoom	hurt the fingerprint itself	unchanged, backup rules' job

And across every edit, on every engine version: not one edit ever produced a wrong film. Edits silence the engine; they don't fool it.

The safety gate. After adding the mirror and bar-trim views I re-ran all 125 real clips: not a single answer changed in either direction. The new views only ever helped edited clips.

The cost. The extra views make the probe about four times slower per clip (about two minutes instead of thirty seconds on my laptop). If this goes towards the pod: normal views first, mirror and bar-trim only when the normal views find nothing, so ordinary clips cost what they cost today.

The one-breath version of each answer

- **What was working before?** The pod's engine: a quick pre-scan, a one-frame probe,

then the rules. It got 52 of my 125 test clips right and 7 wrong.

- **How was it working?** The quick parts decide most clips on counting and best

distance alone; the careful checks live in rules that rarely decide. Four of the seven wrong answers came from the pre-scan, where no rule change can reach.

- **What did I build?** Testing that names the rule behind every answer; a probe

rebuilt with the whole-clip view plus the careful checks, especially offset agreement; an edits test; and fixes for the two edits that blinded the engine.

- **How does it work?** Real scenes match the film at one constant offset,

coincidences scatter. The probe only answers when the evidence agrees the way a real scene does, on bars copied from rules the pod already trusts.

- **Results?** 7 wrong down to 1, keeping 48 of 52 correct. Mirror 0/8 to 8/8,

letterbox 1/8 to 6/8, zero changes on real clips, no edit ever caused a wrong film.